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Is the Use of Interpretations Associated With Treatment Outcome? A Systematic Review and Meta-Analytic Answer

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Interpretation is the doyen of therapeutic techniques and has received some of the greatest theoretical and clinical attention. Nevertheless, to date, no meta-analysis has examined its effect on psychotherapy outcome. In the present systematic review and meta-analysis, we examined the association between interpretations and psychotherapy outcome. The meta-analysis was preregistered in PROSPERO. From 13,195 initially identified abstracts, we extracted 16 independent effect sizes from a total of 897 patients. Findings suggest a medium effect size ($r = -.23$, $p = .002$), indicating that the use of interpretations is associated with better outcome. This association remains consistent across types of interpretation measures, assessor perspectives, study quality, and number of sessions. Sensitivity analyses demonstrated the robustness of the results.

Public Health Significance Statement
In the present systematic review and meta-analysis, we examined the association between interpretations and psychotherapy outcome. Interpretations are defined as techniques implemented by therapists to facilitate insight, and they are theoretically expected to be associated with better treatment outcomes. The findings support the merit of interpretations and their potential to contribute to good treatment outcomes when implemented in the manner and amount considered in the current literature.

Keywords: interpretations, transference, psychodynamic treatment, meta-analysis

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The active ingredients that bring about change in treatment have been the focus of heated debate for the past century and remain unclear to this day. According to psychoanalytic theories, the pillar of psychodynamic work is interpretation. Interpretations are defined as therapeutic techniques used by the therapist to raise the patients' awareness and understanding of recurrent maladaptive patterns in their lives (Gabbard, 2009; Summers & Barber, 2010). Using techniques of interpretations, the therapist labels the patient's experiences (i.e., reframes them in a new way) and integrates them (i.e., creates a link between various components of the patient's experience, such as between personal events and current functioning, past events and everyday experiences, emotional and cognitive components, etc.; Antichi et al., 2022; L. H. Levy, 1963). Often, interpretative interventions are designed to uncover or reveal unconscious conflicts underlying a patient's symptoms, and ascribe new meaning, reason, or explanation to behaviors, thoughts, or feelings. Though, some contemporary writing dwells on the benefits of focusing on preconscious rather than unconscious material, allowing more experience near interpretations (Høglend & Gabbard, 2012).

The literature has suggested many types of interpretation. One way of categorizing interpretations is by the content being interpreted, such as dreams (aiming to infer the latent from the manifest content of a dream), conflicts (e.g., between different wishes), resistance, and transference (see review by Antichi et al., 2022). Another categorization may consider the type of dynamic or conflict being interpreted, such as intrapersonal versus interpersonal. *Intrapersonal* interpretations relate to different parts of the patient's personality and aim to generate insights into intrapsychic conflicts between parts of the ego (Racker, 1968) or between aspects of the self that have conflicting needs (Bromberg, 2003). Other intrapersonal interpretations relate

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Sigal Zilcha-Mano and John R. Keefe contributed equally to this article. Sigal Zilcha-Mano served as a lead for conceptualization, investigation, and writing–original draft. John R. Keefe served as a lead for formal analysis and methodology and served in a supporting role for conceptualization, data curation, and writing–review and editing. Hadar Fisher served as a lead for data curation and writing–review and editing and served in a supporting role for conceptualization and methodology. Tohar Dolev-Amit served as a lead for data curation and writing–review and editing and served in a supporting role for conceptualization, formal analysis, and methodology. Nirit Veler-Poleg contributed equally to writing–review and editing and served in a supporting role for conceptualization and data curation. Jacques P. Barber served in a supporting role for conceptualization, methodology, and writing–review and editing.

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to intrapsychic structures or processes, like defense mechanisms (see review by Olson et al., 2011). Interpretation of defense mechanisms attempts to bring to awareness ways in which an individual avoids the full experience of particular thoughts, affects, wishes, and modes of relating that are distressing or intolerable (Petraglia et al., 2015). *Interpersonal* interpretations focus on the individual's repetitive interpersonal patterns and their possible associations with current suffering, and can be further categorized based on the relationships in the focus of the interpretation. A classic categorization distinguishes between transference and nontransference interpretations (Høglend & Gabbard, 2012). Transference interpretations refer to patient–therapist relations; nontransference interpretation concerns conflicts and interpersonal patterns in the patient's contemporary relationships with others or in memories of past relationships (Høglend & Gabbard, 2012).

Interpretation is one of the psychotherapy constructs that has received the most extensive theoretical and clinical attention, and has critically evolved and changed over the years. Many syntheses of this literature have been published (e.g., McWilliams, 2011; Summers & Barber, 2010). Briefly referring to this extensive literature, some major historical developments can be pointed out: Initially, Breuer and Freud (1895) focused on the recall of significant events of the patient's past through a stream of associations ("free associations"). With the increased interest in dreams as a key to unconscious content, interpretation became a central technique aimed at articulating the analyst's understanding of the patient's hidden or disavowed mental contents (Freud, 1900, 1911). The interpretation could focus on what Freud called the patient's resistance, and when the repressed content is raised to consciousness, interpretations can focus on the defense mechanisms themselves (Racker, 1968). Later, Freud stressed the importance of transference interpretations (Freud, 1912). Freud (1937) also described the analyst's work as archaeology, in the course of which the past is constructed and reconstructed through the use of interpretations. Klein (1948) highlighted the importance of transference interpretations, aiming at exploring unconscious phantasies of the inner world of early and primitive experiences through the patient's experience of the therapist. Winnicott (1962) emphasized that communication between the patient and analyst is based on nonverbal aspects and has an inherent element of reciprocity (1963), and consequently conceptualized interpretations as aimed at "giving back" to patients what they have communicated (1968), serving as a "mirror" for the patient's experiences of self, and avoid interrupting the patient with new imposed insights. According to Bion (1962), analysts use their thinking function to help patients transform emotional experiences into thoughts and become a container for their externalized unacceptable parts (Bion, 1967). In contrast to much of the above, Kohut (1984) argued that empathic interpretations encourage patients to live more harmoniously with their characterizing self-experiences, and stressed the need for therapists to first convey empathic understanding and only then offer a new explanation (Oppenheimer, 2000).

Over the years, the move from one-person to two-person theoretical models of clinical work may have affected the extent to which interpretations are offered in a collaborative manner, in which both the patient and therapist are seen as contributing to the shared experience evolving between them (Mitchell, 2009). Relational psychoanalytic theoreticians have conceptualized therapy as an ongoing intersubjective meeting with the other. In this view of the analytic situation, the therapist and the patient together investigate the patient's experience and reconstruct its meaning (Mitchell & Aron, 1999). Interpretations became a "complex relational event" (Colli et al., 2022), a product of the patient–therapist relationship. Furthermore, "the analytic third" (Ogden, 1994), a contemporary concept, represents the dialectical nature of the unique therapeutic meeting between two subjective individuals intersubjectively affecting each other.

Although originating in the psychodynamic literature, core elements highlighting the importance of bringing mental contents to the patient's awareness can be identified in different manifestations across theoretical orientations (Gazzola et al., 2003; Wampold et al., 2007). For example, in cognitive treatments, identifying and challenging automatic thought distortions and their underlying negative personal schemas (Beck, 1976; Leahy, 2017) has some similarities with psychodynamic interpretations. The same is true regarding schema therapy, in which therapists and patients may engage in identifying and exploring the origin of maladaptive schemas, self-defeating emotional and cognitive patterns established in childhood and repeated through life (Young et al., 2003).

Classically, interpretations are expected to influence outcome by increasing levels of insight, which refers to patients' affectively laden self-understanding of associations between past and present experiences, typical relationship patterns, and the relation between interpersonal challenges, emotional experience, and psychological symptoms. In a recent meta-analysis (Jennissen et al., 2018), insight has been shown to be associated with outcome, but the direct association between interpretations and outcome has not been the subject of any meta-analysis to date.

A recent nonsystematic overview of the association between psychodynamic techniques, assessed directly from therapy process ratings and treatment outcome by Crits-Christoph and Gibbons (2021), suggests that greater use of such techniques was associated with better treatment outcome. Similar suggestions were made in another recent review of the literature, referring to different types of interpretation (e.g., regarding transference, dreams, and symptoms; Antichi et al., 2022). In general, higher levels of psychodynamic interpretations predicted better treatment outcome, but it has been suggested based on the literature that higher levels of transference interpretations predicted poorer treatment outcomes for subsets of patients (Crits-Christoph & Gibbons, 2021). The same was also found in a previous review by Crits-Christoph et al. (2013). In a review of transference interpretations, Høglend (2004) found that a high dose of interpretations may predict a negative process and outcome and does not seem to overcome resistance and defenses. In a general review of the association between outcome and adherence to any technique or competence in it (not limited to interpretations or even to psychodynamic techniques in general), Webb et al. (2010) did not find any significant effect of adherence and competence on outcome. However, Power et al. (2022) looking into the association between adherence, competence, and integrity in any technique found that competence and integrity were significantly associated with outcome.

The present systematic review and meta-analysis aim to expand this review by analyzing the accumulating studies within the framework of a meta-analysis, and exploring potential moderators of the interpretation–outcome association. To achieve our overarching

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goal of testing the effect of interpretations on outcome, we formulated the following subgoals: (a) comprehensively review the literature on the interpretation–outcome association; (b) meta-analyze the literature to quantify the size of the interpretation–outcome association; and (c) search for moderators to explain potential heterogeneity in this association. Based on previous research (Jennissen et al., 2018; Spielmans & Flückiger, 2018), we focused on the following study-level moderators: total rated quality score, adherence to temporal precedence (in which the measurement of interpretations precedes in time the measurement of outcome), study year, setting, study population, number of sessions, type of interpretation measure, and informant (therapist, patient, or external coder).

Clinical Demonstration of the Use of Interpretations

In response to the patient's distress at work, therapists can choose between several types of interpretations, such as intrapersonal, nontransference interpersonal, and transference interpersonal interpretations.

An example of an intrapersonal interpretation:

> You're telling me that your manager disappointed you, you feel hurt, and now you're unsure whether to stay or quit. You feel confused, maybe because you have different and conflicted "voices" in your head: one says to quit to protect yourself from further harm. The other criticizes you for not fighting injustice and condemns you for running away.

An example of a nontransference interpersonal interpretation:

> You're telling me that your manager had disappointed you, and now you're unsure whether to stay or quit your work. You feel hurt and can't see any solution other than quitting. This situation reminds me of when you left home to escape from your father's behavior. Then, you told me you were so helpless and angry that you couldn't believe any change would happen. I wonder if you tend to escape situations where you feel you haven't control over them.

An example of a transference interpretation:

> You're telling me that your manager disappointed you, and now you're unsure whether to stay or quit. Maybe you are also speaking here about how hard it is to be in a relationship where you have expectations from someone to be on your side, but then this individual disappoints you. Thinking more about this, I can see you're uncomfortable now, not looking at me as you usually do. It makes me think about what happened between us last week, when I canceled our meeting at the last moment. Maybe you're also telling me something about us and how you feel about me. Maybe you felt that I also disappointed you. You mentioned earlier that you even considered not coming today.

Method

This meta-analysis was conducted in adherence with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, and was preregistered at the international prospective register of systematic reviews (PROSPERO, registration number: CRD42021255798).

Search Strategy

Eligible studies were quantitative reports of psychotherapy outcome in samples of psychotherapy-seeking adults (see Figure 1). The studies were found in articles reviewing the literature (Crits-Christoph et al., 2013; Crits-Christoph & Gibbons, 2021) and in PubMed and PsycINFO and MEDLINE searches, starting from 1970 to May 2021. These searches used the intersections of the terms: psychotherapy or therapy or treatment, interpretation or adherence or intervention or technique, psychodynamic or psychoanalytic or dynamic or supportive-expressive or insight-oriented, and psychotherapist or therapist. We also contacted 10 experts in the field of interpretation process–outcome research to inquire whether they were aware of any further peer-reviewed studies that we may have missed.

Selection Criteria

The criteria for inclusion were as follows: (a) the authors referred to the therapy process variable as interpretations, psychodynamic adherence, or psychodynamic technique/intervention; (b) the authors provided data on outcome measures; (c) the treatment was individual psychotherapy, rather than group, couple, or family therapy; (d) at least five participants were included in the analysis; (e) the data reported were such that a value indicating the relation between interpretations and outcome could be extracted or estimated; (f) the clients were adults (age >18 years); (g) reports were written in English; and (h) given the documented potential biases in unpublished studies, only studies that were published in a peer-reviewed journal were included to ensure a reasonable quality (Egger et al., 2003; Schmucker et al., 2017).

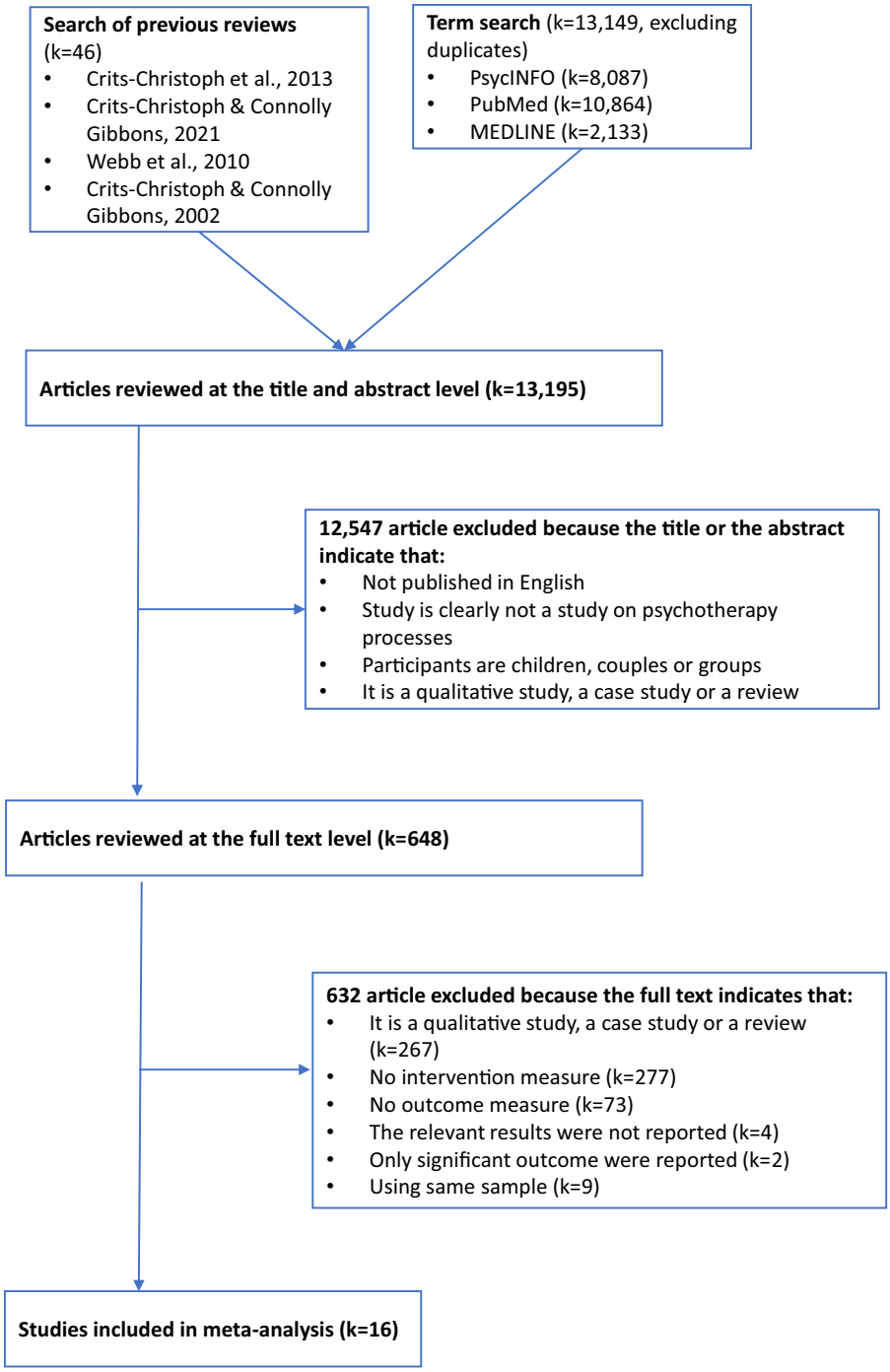
Data Analysis

We used the R package robumeta (Z. Fisher & Tipton, 2015; Pustejovsky & Tipton, 2018), implementing a multilevel meta-analysis approach, to estimate the overall effect size and to explore predictors of effect size using meta-regression. This package conducts robust variance estimation (RVE) meta-regression using both large and small sample RVE estimators under various weighting schemes. These methods are distribution-free and provide valid point estimates, standard errors, and hypothesis tests even when the degree and structure of dependence between effect sizes are unknown. This makes it possible to use all effect estimates available from a given study. We characterized the proportion of between-study heterogeneity in effect sizes using the I^2 statistic (Higgins & Thompson, 2002), with 25% reflecting low heterogeneity, 50% moderate heterogeneity, and 75% high heterogeneity. The I^2 statistic indicates the extent to which between-study differences in effect sizes are not attributable to within-study sampling error. Given that formal tests of heterogeneity are substantively underpowered, we focused on interpreting the size of I^2 versus testing for the presence of heterogeneity (Huedo-Medina et al., 2006). We used the package metafor (Viechtbauer, 2010) to examine publication bias, leave-one-out study sensitivity, and fail-safe N for our results. To explore common patterns of effect sizes and sample size distributions that may indicate the presence of publication bias, both Egger's regression test (Egger et al., 1997) and Henmi and Copas's method (Henmi & Copas, 2010) of detecting possible publication bias were employed. Rosenberg's variant of the fail-safe N was calculated (Rosenberg, 2005), representing the number of studies averaging a null result for the interpretation–outcome correlation that would have to be added to the set of effect sizes to reduce the estimate of the weighted average outcome to nonsignificant level.

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Figure 1

Summary of Studies Included in the Meta-Analysis of Interpretation and Treatment Outcome



Note. See the online article for the color version of this figure.

Risk-of-Bias Appraisal

The methodological quality of the included studies was assessed by two independent reviewers (Hadar Fisher and Tohar Dolev-Amit) according to four categories of the study design: sample, quality of measurements, quality of statistical analyses, and reporting of data (adapted from Jennissen et al., 2018). Any disagreement was discussed by the two reviewers, together with the first and second author, until an agreement was reached.

Power Analysis

We estimated the possible effect sizes that our meta-analysis was adequately powered (≥80%) to detect under different true effect

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sizes and conditions of between-study heterogeneity of effects (Valentine et al., 2010). The average sample size of the 16 included studies was 58.1 participants. Under conditions of low ($I^2 = 25\%$), medium ($I^2 = 50\%$), and high ($I^2 = 75\%$) heterogeneity, we estimated that we had more than adequate (>99%) power to detect medium ($r = .30$) and large ($r = .50$) effects. For small ($r = .10$) effects, we estimated inadequate power to detect possible relationships between interpretation and outcome (low heterogeneity = 75% power; medium = 57% power; high = 33% power).

Results

Population Characteristics

Sixteen studies comprising 897 patients were included in the meta-analysis (see Table 1 for demographic characteristics of the samples and types of interpretations, and Table 2 for study characteristics).

Measures Characteristics

Interpretations

Some of the available measures are not limited only to interpretations, but rather refer to adherence and competence in implementing psychodynamic techniques more broadly defined. Treatment adherence can be defined as the degree to which treatments are delivered with fidelity, as intended, according to the given approach (Fonagy & Luyten, 2019; Owen & Hilsenroth, 2014). A review of the items included in each measure reveals that the vast majority focus on interpretations (e.g., from the multitheoretical list of therapeutic interventions [MULTI]: "The therapist made connections between the client's current situation and his/her past"; from the Penn adherence-competence scale [PACS]: "The therapist focuses attention on similarities among the patient's past and present relationships"; from comparative psychotherapy process scale [CPPS]: "Suggests alternative understanding not previously recognized by the patient"). The fact that the available measures include mainly interpretation-based items may be a product of psychodynamic literature placing interpretation high in the hierarchy of psychodynamic techniques. Given this limitation in the literature, the present meta-analysis does not distinguish between interpretations and psychodynamic techniques more broadly defined, other than for the CPPS, for which a distinction was made between Item 13 (interpretation) and the rest of the scale (S. R. Levy et al., 2015). However, we examined the different measures used as potential moderators of effect sizes.

Measures for assessing interpretations vary between the studies included in the meta-analysis, with the majority of the studies (87.5%) assessing the frequency and intensity of interpretations used in a session. Other studies (6.3%) assessed the competence with which the therapists used the interpretations (Barber et al., 1996), or the accuracy and quality of the interpretations (Crits-Christoph et al., 1988) (6.3%). The specific perspective of the assessor also varies, with 68.8% focusing on the perspective of an external observer (Barber et al., 1996; Connolly Gibbons et al., 2012; Crits-Christoph et al., 1988; Goldman & Gregory, 2009; Hill et al., 1988; Keefe et al., 2019; S. R. Levy et al., 2015; McCarthy et al., 2016; Milbrath et al., 1999; Piper et al., 1991; Tschuschke et al., 2015), 18.8% on the therapists' perspective (H. Fisher et al., 2020; Hendriksen et al., 2011; Jacobs & Warner, 1981), and 12.5% on the patients' perspective (Bush & Meehan, 2011; Glock et al., 2018). The potential validity of patients' perspectives depends on the extent to which they are capable of recognizing particular technical processes occurring within their psychotherapy (DeFife et al., 2008) and their internal scaling of the extent to which such interpretations were used. These abilities and tendencies may be affected by a variety of factors, for example, personality disorders (Lipsitz-Odess et al., 2021).

Treatment Outcome

Measuring treatment outcome, 31.3% of the studies included in the meta-analysis used the severity of depression as the outcome (Barber et al., 1996; Connolly Gibbons et al., 2012; Hendriksen et al., 2011; Hill et al., 1988; McCarthy et al., 2016), 31.25% used general symptom severity (Jacobs & Warner, 1981; S. R. Levy et al., 2015; Milbrath et al., 1999; Piper et al., 1991; Tschuschke et al., 2015), and 31.3% used general life satisfaction and level of functioning or problem improvement as the outcome (Bush & Meehan, 2011; Crits-Christoph et al., 1988; H. Fisher et al., 2020; Glock et al., 2018; Piper et al., 1991). Additional treatment outcomes were panic disorder severity (Keefe et al., 2019), borderline disorder severity, and anxiety symptom severity (Hill et al., 1988), with a single article examining each of these treatment outcomes. Two studies included several outcome measures. The perspective of the outcome assessor also varies, with 62.5% focusing on the patients' perspective (Barber et al., 1996; Bush & Meehan, 2011; H. Fisher et al., 2020; Glock et al., 2018; Goldman & Gregory, 2009; Hill et al., 1988; Jacobs & Warner, 1981; S. R. Levy et al., 2015; Milbrath et al., 1999; Tschuschke et al., 2015) and 18.75% on the external diagnostician's (Hendriksen et al., 2011; Keefe et al., 2019; McCarthy et al., 2016). One study used all three perspectives (Crits-Christoph et al., 1988), another used both patients' and therapists' perspectives (Piper et al., 1991), and yet another study used both external raters' and therapists' perspectives (Connolly Gibbons et al., 2012).

Study design and assessment regime. Most studies of the effect of interpretations on outcome used observers who rated therapists' interventions during one or two sessions, most commonly sampled from the early sessions of the treatment (mainly Sessions 3 or 4). These studies investigated whether interpretations, as implemented mostly early in treatment, predicted symptom change, from pre- to posttreatment (Connolly Gibbons et al., 2012; Crits-Christoph et al., 1988; S. R. Levy et al., 2015). To establish correct temporal precedence between interpretations as a predictor and treatment outcome, several studies used symptom change from the rated session to posttreatment as the outcome (Barber et al., 1996; Keefe et al., 2019; McCarthy et al., 2016). Some of the studies have also estimated the contribution of the therapist to the variance in the interpretation–outcome association (S. R. Levy et al., 2015).

Recently, there has been a general shift in psychotherapy research to focusing on within-treatment processes as they unfold from one session to the next within each patient (Barkham et al., 2021), which is starting to affect the research on interpretation as well. Given the labor intensiveness of coding several sessions per individual for a substantial sample of individuals, these studies used mainly patient and therapist self-report questionnaires to test the association between interpretations at time T and outcome at the following session (time $T + 1$) (Boswell et al., 2010; H. Fisher et al., 2020). Their objective was to determine whether greater use of interpretation

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Table 1
Demographic Characteristics of the Samples and Types of Interpretations in the Studies Included in the Meta-Analysis

Study	Age	Gender	Marital status	Racial/ethnic identification	Education (years)	Employment	Type of interpretation
Barber et al. (1996)	Mean: 38.4 ($SD = 10.36$)	72% Female	N/A	N/A	N/A	N/A	N/A
Bush and Meehan (2011)[a]	Mean: 59	62.9% Male, 34.8% Female 2.24% missing data	—	—	—	—	N/A
Connolly Gibbons et al. (2012)	Mean: 40.79 ($SD = 10.35$)	90% Female	46.2% Single	15.4% White, 84.6% African American	Mean: 11.9 ($SD = 1.86$)	65.8% Unemployed	N/A
Crits-Christoph et al. (1988)	65.11% between ages 15 and 24	69.76% Female	67.44% Single	N/A	48.83% Some college degree	N/A	N/A
H. Fisher et al. (2020)	Mean: 40 ($SD = 13.7$)	58% Female	43% Single	N/A	59% At least a bachelor's degree	82% Employed	N/A
Glock et al. (2018)	Mean: 26.7 ($SD = 10.7$)	79.41% Female	N/A	85.4% White, 5.9% Native American, 2.9% Asian American, 2.9% African American, 2.9% Latinx	N/A	N/A	Nontransference interpretation
Goldman and Gregory (2009)	Mean: 27.40 ($SD = 6.85$)	90% Female	70% Single	N/A	Mean: 13.50 ($SD = 3.06$)	N/A	N/A
Hendriksen et al. (2011)	70% Younger than 39 years	70% Female	N/A	N/A	50% Highly educated	N/A	N/A
Hill et al. (1988)	Mean: 42.38 ($SD = 9.41$)	100% Female	N/A	N/A	N/A	N/A	N/A
Jacobs and Warner (1981)	Median: 29 (range 13–73)	66.67% Female	26% Single	99.9% White	37% Less than high school education	N/A	N/A
Keefe et al. (2019)	Mean: 39.5 ($SD = 14$)	67.7% Female	N/A	83.1% White, 16.9% African American, 12.3% Latinx	N/A	N/A	N/A
S. R. Levy et al. (2015)	Mean: 29.68 ($SD = 10$)	72.37% Female	60.53% Single	N/A	N/A	N/A	Nontransference interpretation
McCarthy et al. (2016)	Mean: 35.5 ($SD = 12$)	61% Female	64% Single	45% White, 49% African American, 6% Asian American, 3% Latinx	Mean: 14.3 ($SD = 2.2$)	36% Unemployed	N/A
Milbrath et al. (1999)	Mean: 40.2 (range 26–52)	100% Female	N/A	80% White, 5% Latinx, 15% did not report their ethnicity	The average educational attainment for the group was a high school education	N/A	Nontransference interpretation
Piper et al. (1991)	Mean: 32	62% Female	N/A	N/A	N/A	N/A	Transference interpretation
Tschuschke et al. (2015)	Mean: 39.6 ($SD = 11.8$)	56.79% Female	51.9% Single	N/A	40.7% apprenticeship certificate 19.8% high school diploma	N/A	N/A

[a] In this study, the therapists rated the interventions.

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Table 2

Summary of Studies Included in the Meta-Analysis of Interpretation and Treatment Outcome

Publication	Country	Treatment type	RCT yes/no	Disorder	Interpretations			Outcome			Effect size	
					Measure	Rater	Time	Measure	Rater	Time	r^a	n
Barber et al. (1996)	United States	SET	No	MDD	PACS	O	Session 3	BDI	P	Pre and post	−.34	29
Bush and Meehan (2011)	United States	Psychoanalytic	No	Various	Classic technique	P	Session 1	Global outcome questions	P	Session 1	−.43	89
Connolly Gibbons et al. (2012)	United States	SET/TAU	Yes	Moderate to severe depression	PACS	O	Session 3	HRSD	O	Each session	−.47	40
								BASIS-24	P			
Crits-Christoph et al. (1988)	United States	PP	No	Various	CCRT	O	Two early sessions	Residual gain	P/O	Pre and post	−.25	43
								Rated benefits	P/T			
H. Fisher et al. (2020)	Israel	STPP	No	Various	MULTI-30	T	Each session	ORS	P	Each session	−.04	122
Glock et al. (2018)	United States	Various	No	Anxiety	CPPS, item #13	P	Unknown	Overall improvement	P	Unknown	−.32	34
Goldman and Gregory (2009)	United States	DDP	No	Borderline PD and alcohol use	Adherence to DDP	O	Pre, 3 months, 6 months, 9 months, and 12 months	BEST	P	Pre, 3 months, 6 months, 9 months, and 12 months	−.64	10
Hendriksen et al. (2011)	Netherlands	Psychotherapy/combined therapy	Yes	MDD	TEF	T	Unknown	HRSD	O	Pre and post	−.29	147
Hill et al. (1988)	United States	Various	No	Women over 25 years of age with self-esteem and relationship problems	Counselor verbal response modes category system	O		SCL-R anxiety scale	P	Pre and post	−.36	8
								SCL-R depression scale	P			
								TSCS	P			
Jacobs and Warner (1981)	United States	PP	No	Various	Staff attitudes and therapeutic style	T	First 2 weeks	Global improvement scale	P	Unknown	−.02	161
Keefe et al. (2019)	United States	PFPP	Yes	Panic Disorder	PFPP-RS	O	Sessions 2 and 10	PDSS	O	Pre, Week 1, Week 5, Week 9, post	−.37	65
S. R. Levy et al. (2015)	United States	STPP	No	Various	CPPS, item #13	O	Sessions 3 and 9	GSI-RCI	P	Pre and post	−.33	76
McCarthy et al. (2016)	United States	PP/pharmacotherapy/placebo	Yes	MDD	MULTI	O	Week 4	HRSD	O	Pre, Week 4, and post	−.02	33
Milbrath et al. (1999)	Canada	Brief dynamic psychotherapy for stress response symptoms	No	Bereaved subjects	PIRS	O	Session 4	BSI	P	Pre and post	−.48	20
Piper et al. (1991)	Canada	PP	Yes	Various	TIRS	O	Sessions 4, 7, 9, 11, 14, 16, 18, and 20	General symptoms and dysfunction	P	Pre, post, and 5-month follow-up	.29	64
								Overall usefulness of therapy	P/T			
								Sexual and social functioning subscale	P			
								Target objective severity	T/Family			
Tschuschke et al. (2015)	Switzerland	Various	No	Various	Treatment adherence rating	O	Early, middle, and late	Combined OQ-45, GSI–BSI, GAF	P	Pre and post	.008	81

Note. Treatment type: STPP = short-term psychodynamic psychotherapy; PFPP = panic-focused psychodynamic psychotherapy; PP = psychodynamic psychotherapy; SET = supportive-expressive treatment; TAU = treatment-as-usual; DDP = dynamic deconstructive psychotherapy. Design: RCT = randomized controlled trial. Disorder: PD = personality disorders; MDD = major depressive disorder. Interpretation measure: MULTI = multitheoretical list of therapeutic interventions; TSCS = Tennessee self-concept scale; PFPP-RS = panic-focused psychodynamic psychotherapy rating scale; CPPS = comparative psychotherapy process scale; PACS = Penn adherence/competence scale; TEF = therapist evaluation form; PIRS = psychodynamic intervention rating scale; TIRS = therapist intervention rating system; CCRT = core conflictual relationship theme. Rater: T = therapist; P = patient; O = observer. Outcome measure: ORS = outcome rating scale; PDSS = panic disorder severity scale; HRSD = Hamilton rating scale for depression; GSI-RCI = global severity index-reliable change index; BASIS-24=behavior and symptom identification scale; BEST = borderline evaluation of severity over time; BSI = brief symptom inventory; BDI = beck depression inventory; SCL-R = symptom check list-revised; OQ = outcome questionnaire; GAF = global assessment of functioning.

[a] Average *r*, with some studies contributing multiple effects. We used a method that made it possible to include multiple effects from the same article.

resulted in a greater reduction in symptoms at the subsequent session. Although not included in the meta-analysis because of the comparison of packages of treatments rather than using specific techniques, another line of research uses randomized controlled trials (RCTs) to establish a causal relation between techniques and outcome (Høglend, 1993; Høglend et al., 2008; Zilcha-Mano et al., 2021).

Overall Meta-Analysis and Heterogeneity

Higher scores on interpretation measures were significantly associated with more symptom improvement, $r = -.23$, 95% CI: [−0.36, −0.11], $t(12.9) = -3.95$, $p = .002$, albeit with a medium degree of between-study heterogeneity ($I^2 = 58.3\%$).[1] A forest plot of the interpretation effect size for each study and the weighted average effect size can be viewed in Figure 2. When examining the sensitivity of the result to the inclusion of single studies, the removal of the Jacobs and Warner (1981) study lowered between-study heterogeneity, $r = -.26$, [−0.38, −0.15], $t(11.1) = -4.91$, $p < .001$, $I^2 = 42.9\%$, indicating it may be a possible outlier. There was no indication of publication bias per Egger's regression test ($t = -.51$, $p = .617$, intercept = −0.19, [−0.47, 0.10]; see Figures S1 and S2 in the online supplementary material for regular and contour-enhanced funnel plots) or Henmi and Copas's method (r difference < .01). Rosenberg's fail-safe n was estimated at 149 (Rosenberg, 2005).

Moderators

We examined several possible study-level moderators of the relationship between interpretations and outcomes that could explain heterogeneity between studies. The total rated quality score did not predictably relate to effect sizes, $B = -0.00$, 95% CI: [−0.06, 0.05], $t(6.9) = -0.15$, $p = .885$, and studies strictly adhering to temporal precedence (i.e., that interpretation could only predict change after the process measurement) did not have reliably different effects, $B = 0.05$, [−0.26, 0.35], $t(5.4) = 0.38$, $p = .719$. Nor was study year related to effect sizes, $B = -0.01$, [−0.02, 0.01], $t(4.9) = -1.39$, $p = .223$. RCTs did not have distinguishably different effect sizes, $B = 0.01$, [−0.29, 0.32], $t(9.5) = 0.11$, $p = .915$. Studies primarily treating major depressive disorder did not differ from other studies, $B = -0.09$, [−0.38, 0.20], $t(5.4) = -0.79$, $p = .466$. Studies with a higher number of sessions did not have a significantly different effect size, $B = -0.01$, [−0.01, 0.00], $t(2.9) = -1.99$, $p = .146$.

Adherence scores did not significantly differ from competence scores in predicting outcomes, competence $B = -0.12$, 95% CI: [−0.45, 0.22], $t(2.4) = -1.27$, $p = .311$. Use of neither observer report, $B = -0.05$, [−0.33, 0.23], $t(9.2) = -0.41$, $p = .489$, nor therapist report, $B = 0.14$, [−1.11, 1.40], $t(1.5) = 0.70$, $p = .577$, measures of interpretation affected effect sizes. Use of the MULTI, $B = 0.23$, [−0.23, 0.69], $t(1.4) = 3.48$, $p = .125$, or a core conflictual relationship theme, $B = -0.15$, [−0.45, 0.16], $t(3.0) = -1.54$, $p = .221$, measure of interpretation did not result in different effect estimates. Insufficient studies were available to meta-analytically examine the role of different focus of interpretation (e.g., transference; $k = 1$, while remaining studies generally did not distinguish interpretation focus or type) or potential moderators ($k = 4$; all different constructs) or mediators ($k = 1$) of the interpretation–outcome relationship.

Discussion

This meta-analysis, based on the data of 897 patients from 16 independent samples, suggests that the use of interpretations is associated with better treatment outcome. This association is low-to-moderate in size, and is comparable to other key mechanisms in treatment, such as the therapeutic alliance (Eubanks et al., 2018). The association remains consistent across all the moderators examined, including types of interpretation measure (adherence vs. competence), assessor perspectives, study quality, whether temporal separation between interpretations and subsequent outcomes was established, and number of sessions. The null results of formal publication bias tests (Egger's test; the Henmi and Copas method) and a high fail-safe n (Rosenberg, 2005), indicating that a large number of (on average) null studies would have to be added to our data to invalidate our finding suggest that these results are also potentially robust against file drawer effects (i.e., missing potential studies with null or negative findings).

The significant effect of interpretations on outcome is consistent with the general effect of technique competence on outcome, as documented in the meta-analysis by Power et al. (2022). The results are in contrast to the general null effect of technique adherence and competence on outcome, as documented in the meta-analysis by Webb et al. (2010). The findings are also in contrast to a previous RCT that manipulated the use of transference interpretations and found that use of transference interpretations was not related to outcome generally (Høglend et al., 2006), although they were related to better short- and long-term (4 years) outcomes for patients with poor object relations or personality disorder (Høglend et al., 2008, 2011). A possible explanation of this contrast is a possible treatment-specific nature of the effects of techniques, stressing the need to disentangle the effect of each type of technique on outcome. For example, interpretations referring to the here-and-now therapeutic relationship between the therapist and patient can be curative in treatment that focuses on identifying maladaptive patterns of interpersonal relationships, but may be counterproductive when used as a defensive response by the therapist in reaction to the patient's challenging of behavioral tasks used in cognitive-behavioral treatment. It is also possible that focusing more specifically on interpretations as compared to general psychodynamic adherence (as in Webb et al., 2010) reveals a different pattern of findings; for example, in one study examining the relationship of psychodynamic interventions to subsequent outcomes in the treatment of panic disorder, greater use of panic-focused interpretations was associated with greater panic improvements, while clarifications and nonpanic focused interpretations were not (Keefe et al., 2019).

Limitation and Further Research

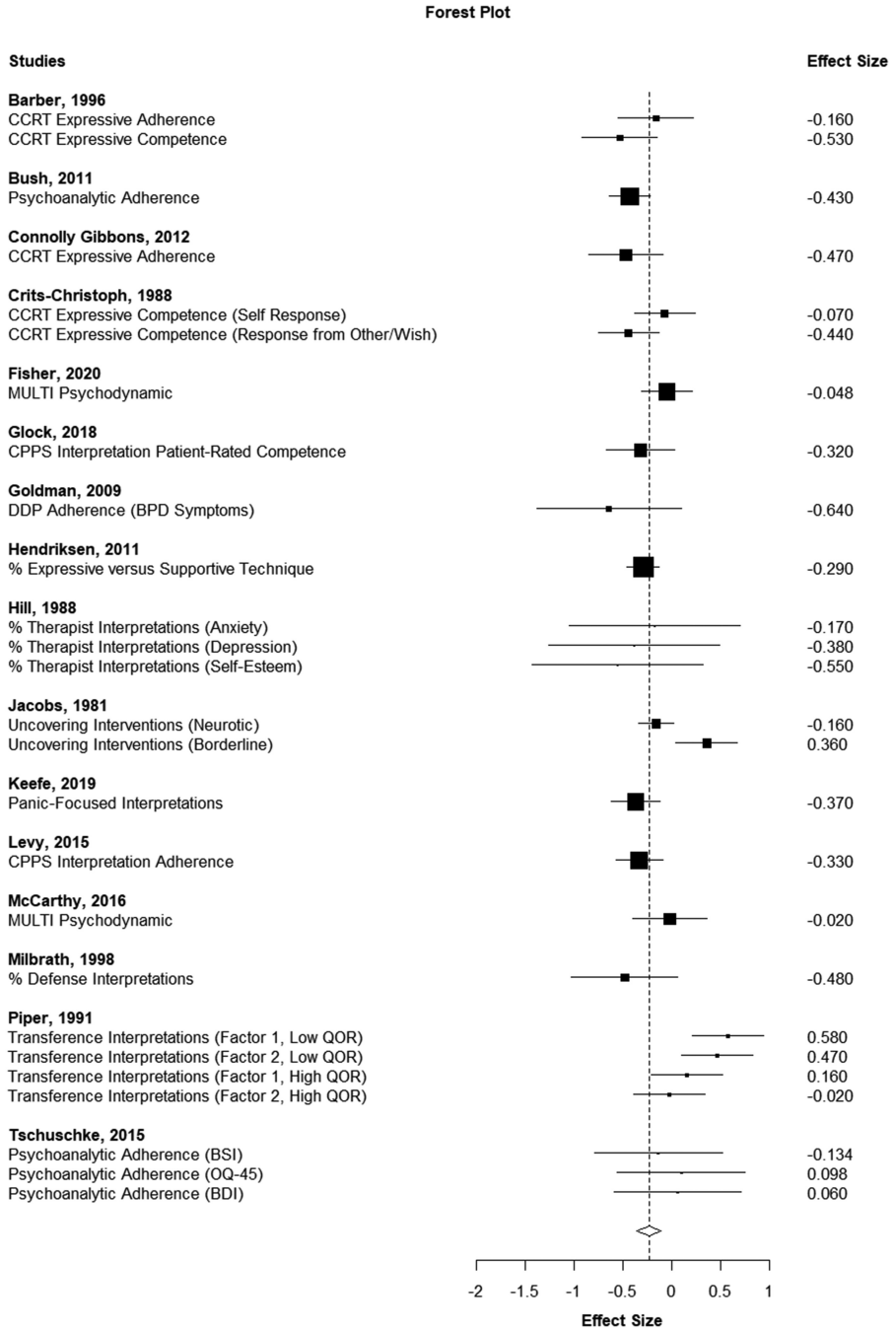
Despite the strong emphasis on the importance of interpretation, empirical studies in this area have been more limited than one may

[1] During the review process, a reviewer requested that we provide confidence intervals for the I^2 index of between-study heterogeneity. As our preregistered analytic method *robumeta* does not calculate confidence intervals for the I^2 index, we employed the function "rma.mv" in the *metafor* package (Viechtbauer, 2010), which conducts a multilevel meta-analysis using some different computational principles from *robumeta*. Nevertheless, using this alternative analytic technique, we arrived at a very similar finding for the interpretation-outcome correlation ($r = -.22$, 95% CI [−0.34, −0.10], $SE = 0.06$, $Z = -3.58$, $p = .0003$) and between-study heterogeneity ($r =$ 56.5%, [30.3, 79.0]).

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Figure 2

Forest Plot Showing the Meta-Analysis Results for the Effect of Interpretation on Treatment Outcome



Note. Negative effect means that interpretation was associated with symptoms decrease.

expect for such a key construct, thereby limiting the sample size on which this meta-analysis is based. Furthermore, moderate heterogeneity ($I^2 = 58.3\%$) emerged in the interpretation–outcome correlation between studies, in other words, although we detected an overall positive interpretation–outcome correlation, the detected level of heterogeneity suggests that true differences in interpretation–outcome correlations exist between studies. This points to the need to explore in greater depth for whom interpretations should be more intensely used, by whom, and when.

The lack of significant study-level moderators can be a product of the small sample size rather than the lack of potential factors moderating this association, which may not be at the study level (e.g.,

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whether the study treated depression) but rather at the level of the patient, therapist, or dyad. The limited number of studies testing moderators and the fact that each focused on different ones precluded meta-analyzing within-study moderators. The great diversity between studies in the measures used to evaluate interpretations, in patient and therapist populations, and in study designs contributed to the generalizability of the findings but further impaired its specificity and may have contributed to heterogeneity. Future studies should explore patients' characteristics (e.g., object relations or interpersonal problems; Høglend et al., 2006; Keefe et al., 2019), therapists' characteristics (e.g., training, clinical experience), researchers' allegiance, timing within the treatment and the session (Keefe et al., 2019), the quality and nature of the interpretation (e.g., accuracy; Crits-Christoph et al., 1988; Silberschatz et al., 1986), affect awareness (Høglend & Hagtvet, 2019), and the effect of responsiveness as potential moderators of the effect of interpretations on outcome. Regarding the therapeutic alliance between patient and therapist, it was previously found that more psychodynamic techniques were associated with better postsession changes under conditions of higher alliances (Owen et al., 2013). In another study, a curvilinear relation was found, specifically, a moderate level of counselor adherence was associated with better outcome in the context of a low alliance (Barber et al., 2006). However, these could not be included in the meta-analysis because they did not examine the direct association between interpretations and outcome, and too few studies explored the association of interpretations and alliance with outcome to examine in a meta-analysis.

Other limitations include the correlational nature of the association, which precludes inferring causality. For example, the level of object relation or interpersonal capacities may serve as a third variable affecting both interpretation and outcome (Høglend et al., 2006). RCTs manipulating the use of interpretations are critical in this regard, suggesting the merit of interpretations, especially for a subset of individuals with more impaired interpersonal capabilities (Høglend et al., 2006; Zilcha-Mano et al., 2021). Moreover, only one correlational study (H. Fisher et al., 2020) reported on and identified on a potential mediator (self-understanding) of the interpretation–outcome relationship. This finding is convergent with results from the Høglend trial (2006, 2008) manipulating the provision of transference interpretations in psychodynamic therapy suggesting that, for patients for whom transference interpretations were helpful, this advantage is partially mediated by greater improvements in insight (Johansson et al., 2010).

Future studies should also focus on the within-individual association between interpretations and outcome to test whether the use of techniques in a given session predicts treatment outcome in the subsequent session repeatedly over the course of treatment (Zilcha-Mano, 2021). Such within-individual studies should also test within-individual mediation models examining whether insight is the mechanism mediating the effect of interpretations on outcome. It is important to stress that the findings refer to the amount and timing of interpretations used in the studies included in the meta-analysis. Both high and low use may result in less beneficial effects than a moderate level of interpretation (McCarthy et al., 2016). In the present study, we could not examine a curvilinear effect of interpretations on outcome because there were not enough studies that tested this association, and not enough studies that used the same measure to assess between-studies mean interpretation level as a potential moderator.

The findings of this meta-analysis that the use of interpretations is associated with better treatment outcomes raise many further questions about how interpretations can be effectively used and tailored to the individual patient. Theoretical and clinical writing, as well as case studies, provides many hypotheses to be tested in future empirical research. For example, it has been suggested that interpretations, especially transference interpretations, which focus on the recurrence of the patient's maladaptive patterns in the therapeutic relationship, should be carefully implemented because this high-risk high-gain technique (Gabbard, 2006) may have a negative effect on the strength of the alliance (Piper et al., 1991) and on treatment success (Ogrodniczuk et al., 1999; Schut et al., 2005). These findings should be interpreted with caution, however, because therapists may increase their use of transference interventions with more challenging patients in the case of a negative or hostile therapeutic process (Colli et al., 2022; Høglend, 2014). Similarly, it has been suggested that interpretations should be used when the alliance between patient and therapist is strong (Antichi et al., 2022; Owen & Hilsenroth, 2014), and be offered in manners that empower the patients (Wachtel, 2011). For example, one case study demonstrated how the therapist identified a therapeutic change in the patient's behavior patterns and delivered an interpretation that "celebrated" it, encouraging the patient to take further steps in an adaptive direction (Leibovich et al., 2020).

As demonstrated in several case studies, effective use of interpretations may include substantive use of supportive techniques (Cohen et al., 2019; Van Nieuwenhove et al., 2020). Similarly, affirmative appreciation of the patient's internal experience (Høglend & Gabbard, 2012) and attributing the insight to the patient rather than delivering it from outside (Farber et al., 2019) are some of the other conditions that may contribute to effective interpretations. For example, a case study reported that when a therapist working with a young woman validated a rupture in the alliance, assumed responsibility for it, and asked the patient about her feelings and understanding, a meaningful therapeutic breakthrough occurred (Leibovich et al., 2020).

To offer interpretations in a sensitive, responsive, and tailored manner, most effective for a given patient (Tummala-Narra, 2016), it is important to collect detailed data about the patients, including gender (not only as a binary variable), immigration, religion, sexual orientation, ethnicity, and race (Watkins, 2013). Diversity within study cohorts is crucial to ensure future progress in tailoring the use of interpretations to the specific characteristics and needs of a given patient. Future studies should add sociodemographic and cultural variables as potential moderators to shed new light on how the use of techniques can be tailored to the individual to improve treatment efficacy. Studies are also needed on the extent to which patients experience the interpretation as helpful, and are interested and able to integrate it in their perception and understanding of the self and other. Similarly, more research is needed on therapists' responsiveness to patients' reactions (verbal and nonverbal) when providing an interpretation (Watson & Wiseman, 2021). Other issues awaiting future studies are the types of interpretations that are most effective with individual patients in particular circumstances, the time in treatment relative to the full treatment length when the interpretations are provided, the type and amount of work accomplished until that point in treatment, and differences between short-term and long-term treatments. Research is also needed on how additional mechanisms of change, within

psychodynamic literature (e.g., expressing and experiencing affect; Aafjes-van Doorn & Barber, 2017) and outside of it are dynamically related to the provision and effect of interpretations.

Conclusion

To the best of our knowledge, this is the first meta-analysis to test the association between interpretations and treatment outcome. The findings suggest that the use of interpretations is associated with better treatment outcomes. A medium level of heterogeneity characterizes this effect, suggesting that some individuals in some situations may benefit from interpretations more than others. Further research should help to elucidate the therapeutic conditions under which interpretations may be more likely to be helpful, patient or therapist characteristics influencing the impact of interpretations, experimental manipulation of interpretations, and on potential mediators of the interpretation–outcome relationship.

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